

Security, Error Handling and Test Design

Project Design Phase | Import Data Using Transform Maps

1. Security Design

Area	Design
Roles	admin or import_transformer required to load data and run transforms
Table access	ACLs restrict who can read staging and target tables
Personal data	Files with personal data are handled per policy and deleted after import when required
Script safety	Only trusted admins edit transform scripts; changes tracked in update sets
Environment	Imports tested in sub-production before production

2. Validation Design

Check	When	Method
File format and headers	Before upload	Template with fixed headers
Required fields present	onBefore	Script ignores incomplete rows
Unique key present	Coalesce	Rows without key are ignored
Reference exists	Field map lookup	Reference match; log if missing
Row counts match	After load	Compare file rows with staging rows

3. Error Handling Design

Error type	Example	Handling	Where to see it
Data error	Missing email	Row ignored	Transform history
Reference error	Unknown department	Warning, field left blank	Import log
Mapping error	Wrong target field	Fix map and re-run	Target table review
Script error	Exception in script	Row marked Error	System log
Format error	Wrong sheet / header row	Fix data source	Load data message

4. Logging and Audit Design

- Import Set records the file, date, user and row states.
- Transform History records each run with counts of inserted, updated, ignored and error rows.
- Script log.info / log.warn messages explain skipped rows.
- Retention follows the organisation's policy.

5. Rollback and Recovery

- Take a backup or update set before first production run.

- Records from a run can be identified through the import set reference.
- Wrong updates are corrected by fixing the file and re-running with the same coalesce key.
- Wrongly inserted records are removed by filtering on the import set.

6. Test Design (carried to Planning phase)

ID	Scenario	Expected result
T-01	Valid 10-row file	10 inserted
T-02	Same file loaded again	10 updated, 0 inserted
T-03	Row without email	Row ignored and logged
T-04	Unknown department	Record saved with blank department, warning logged
T-05	Status values Active / Inactive	active true / false correctly
T-06	User without role runs import	Access denied